



Boundary Role Spanning Behavior, Conflicts and Performance of Industrial Product Managers

Steven Lysonski and Arch G. Woodside

While researchers have probed predictors of product managers' performance in consumer goods firms, few have looked at the performance of the industrial product manager. In this article, Steven Lysonski and Arch Woodside use path analysis to examine causal models of industrial product managers' performance during periods of rapid technological change. Key variables are analyzed, including the effects of environmental uncertainty, boundary spanning behavior, role pressures, role outcomes and two performance measures. A total of 69 industrial product managers from New Zealand Telecom completed a questionnaire that included both operational measures of the key variables and their overall job performance. The results suggest that most of the hypothesized relationships are supportable empirically. Performance was hindered by environmental uncertainty, role conflict, role ambiguity, tension and dissatisfaction. The authors discuss their study's implications for improving the effectiveness of product management practices.

Using the telecommunications industry in a case study, Jelinek [37] describes how technology can radically change the basis for competition. For the telecommunications industry and many others Jelinek emphasizes that the "new technology" based on semiconductors, versus the old technology based on electromechanics, results in the following strategic imperatives:

- custom tailored and individualized products are not only feasible but economical;
- proliferating products and flexible response to customers means more finely tuned market segmentation and product positioning strategies; and
- as more competitors push the state-of-the-art technology and product design, competition will favor those who provide quality, reliability and responsiveness; price alone won't do it, nor will the technology.

Jelinek's 1984 observations were based on her analysis of American Telephone and Telegraph and other US telecommunications businesses. By the late 1980s the strategic imperatives outlined by Jelinek occurred worldwide in market-based nations. The product proliferation of high quality electronic telecommunications products coupled with customers more frequently demanding many product features, more speed and higher reliability has resulted in major reorganizations of marketing and related business activities within large telecommunications companies.

The marketing effectiveness following the reorganization response to the new technology and

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strategic imperatives by a national telecommunications enterprise. New Zealand Telecom, is reported in the present article. In a major effort to successfully introduce many high technology products in a marketplace that included aggressive competitors with many new products of their own, New Zealand Telecom (NZ Telecom) began by creating a new marketing organization. The product manager position product manager was central in the new marketing organization; before the marketing reorganization, the product manager position did not exist. The new product manager job description included the responsibility for learning customer acceptance to new product designs, quickly learning and reporting product market introductions by competitors and directing the implementation of marketing strategies by the operating divisions of the firm. The study reported here is based on an in-depth survey of the activities and performance of these product managers at NZ Telecom.

Success for industrial products may hinge on the performance of the product manager in coordinating the activities of others and in effectively managing the assigned product lines. Yet little is known about the position or the characteristics of individual industrial product managers [19] despite the increasing popularity of the industrial product management concept. Even though a growing literature has developed on the role of the product manager in consumer goods industries [14, 28, 49-51], researchers have not probed the many dimensions of the industrial product manager. No attempts seem to have been made to develop and test models of the antecedents of performance of industrial product management.

The purpose of this study is to develop valid causal models of the performance of the industrial product manager.

The product manager in either the industrial or consumer setting functions as a communicator both inside and outside the firm. This boundary spanning role links the organization with a network of other organizations such as suppliers and buyers. In addition, this role includes coordinating the efforts of multiple corporate departments to achieve product success. This communication role, however, may create stress and conflict for product managers, which ultimately can impair their performances.

Although some research has examined the consumer product manager in terms of boundary spanning behavior [50], no comparable studies have looked at the industrial product manager. This article builds on the foundation laid by previous research on boundary role theory [1, 56] and provides a theoretical and empirical synthesis of some key issues related to the industrial product manager.

The article is divided into four sections. The first reviews the published studies on industrial product management. The second presents the research model and hypotheses tested, while the third develops a methodology for testing the model and hypotheses. The last section focuses on the empirical findings and presents the conclusions and implications.

Literature on Industrial Product Managers

Only a few major studies have examined the industrial product manager and none of these has addressed performance, particularly in terms of the effects of environmental and behavioral dimensions on performance. Such studies have established some differences between industrial and consumer product managers. Kelly and Hise [42], for example, found that industrial product managers were generally older and had more years of experience with the same product compared to product managers in consumer goods. Moreover, they found that in terms of role conflict, role clarity and tension, the only difference was that consumer product managers reported greater role clarity. Although the responsibilities of consumer and industrial product managers did not differ significantly in their study, product

managers with industrial goods responsibilities had greater levels of contact with sales reps, customers and managers in distribution. In another study using the case study approach, Petrini and Grub [63] found that industrial product managers at Borg Warner and Curtis Wright had no control over pricing and no profit and loss responsibility.

Two more recent studies on industrial product management find even greater differences between consumer and industrial product managers than earlier reports. Cummings et al. [19] found that consumer product managers were active decision makers, unlike their industrial counterparts who have less decision-making influence and instead function more as coordinators. They concluded that the industrial product manager differs significantly from the consumer product manager in terms of background characteristics and decision-making authority. Industrial product managers were significantly older, had more work experience and reached their position primarily from sales and some from engineering, unlike their consumer counterparts who were

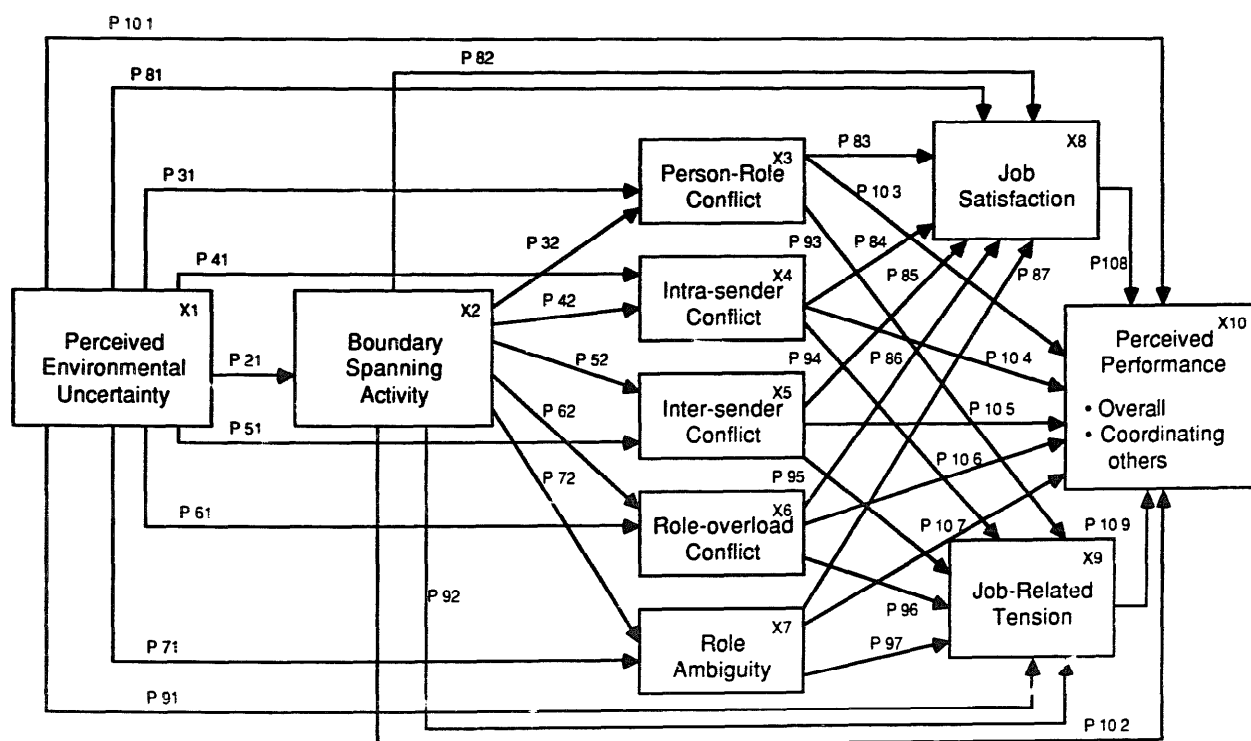
more likely to start as assistant product managers after obtaining M.B.A.s.

The other study by Eckles and Novotny [24] found that industrial product managers had responsibility for six or more products on average. Their major responsibilities include the products' profitability, physical product changes, overseeing the products' progress, product reengineering, market entry, product conceptualization and any product deletions. Given such differences, research findings on consumer product managers may not be as applicable to the industrial setting. Hence, there is a need to focus explicitly on the industrial product manager.

Variables and Hypotheses

The relationships of several key behavioral variables related to the product manager's performance were examined in the present study. The model developed in Figure 1 is a causal one in which perceived environmental uncertainty, boundary spanning and role pressures of conflict and ambiguity were predictors of the product

Figure 1. A causal model of the variables in the study. Perceived environmental uncertainty, boundary spanning, role perceptions of conflict and ambiguity, role outcomes and perceived performance.



manager's job satisfaction and job-related tension. In addition, these several variables were hypothesized to be related to two measures of performance. These variables were selected to encompass the unique characteristics of the industrial product management role and to provide actionable implications for industrial product managers. Such variables have also been found to be important in other settings such as sales management, consumer product management and project management [e.g., 26, 39]. Although the model does not provide a complete explanation of the forces operating on the product manager, it serves to summarize and integrate most of what is known about the relationships featured.

According to Kahn et al. [39, p. 89], "role conflict and ambiguity are independent sources of stress." Hence, the correlation between role ambiguity and conflict is viewed as unrelated. In addition, satisfaction and job-related tension are not linked in the model. High- versus moderate- to low-level job-related tension may be hypothesized to have a negative influence on job satisfaction; however, no significant linear or quadratic relationships between levels of job satisfaction and job-related tension have been reported in previous studies.

The central focus of this study is to explore the hypotheses stemming from the relationships in the model, the rationales for which have been developed in a number of theoretical and empirical work. While the logic of some of these predictions is perhaps self-evident, the predictions are nevertheless important. A brief discussion of each set of linkages depicted in the model is developed, followed by the hypotheses tested.

Hypotheses

Uncertainty, which deals with the perceived predictability of the environment, is a key construct in the model. High perceived uncertainty imposes a difficult environment in which to make decisions. Galbraith [27], for example, stresses that high levels of uncertainty require increased information processing by managers while Thompson [74] views uncertainty as the fundamental problem in complex organizations. Boundary spanning activities, therefore, become instrumental for gathering and integrating information to adapt to uncertain environments [12, 13, 18].

Industrial product managers operating in uncertain market settings may confront turbulences which can threaten the viability of their product line. Such turbulences likely include market introductions of high technology products by aggressive competitors, as well as the responsibility to gain customer acceptance of an enterprise's own high technology product launches. Conversely, stable environments are unlikely to create the same challenge in decision making, coordinating and information gathering. Hence, the frequency of the product manager's boundary spanning activities to collect information, opinions and ideas is likely to be greater in uncertain environments. Expressed as an hypothesis, the following prediction is made:

H1: As industrial product managers perceive greater environmental uncertainty, their boundary spanning activity is greater.

Uncertainty may also impinge on the product manager's role conflict in terms of how others should cooperate with the product manager and on the degree of role ambiguity the product manager experiences in wondering how s/he will be evaluated and compensated. Similarly, perceived environmental uncertainty is likely to affect the product manager's satisfaction and job tension. Finally, the product manager's performance is likely to be adversely affected if the environmental uncertainty s/he faces seems overwhelming. Others have documented the role difficulties experienced in decision making in uncertain environments [9, 39, 68, 50, 77].

In view of these arguments, the following hypotheses are proposed:

H2: An industrial product manager's perceived uncertainty is related:

- H2a: Positively to role conflict dimensions
- H2b: Positively to role ambiguity
- H2c: Negatively to job satisfaction
- H2d: Positively to job-related tension
- H2e: Negatively to performance.

As product managers perform their functions, they may confront diverse and conflicting expectations about how their job should be performed. Moreover, they may have to compete for finite resources by using interpersonal skills in bargaining, compromising or cajoling. Several research-

ers have found boundary spanning to be associated positively with role conflict [39, 61, 64, 76] and also likely to create a feeling of role ambiguity [41]. According to this background, the following hypotheses are suggested:

H3: The frequency of an industrial product manager's boundary spanning is related:

H3a: Positively with role conflict dimensions

H3b: Positively with role ambiguity.

Boundary spanning activity is also postulated to be associated with the product manager's behavioral outcomes such as job satisfaction and job-related tension. One study has found some association between boundary spanning and consumer product manager role outcomes [50]. In addition, because boundary spanning is an important activity for integrating and coordinating a product line, it seems plausible that increased frequency of boundary spanning will be associated with performance. Given these rationales, the following hypotheses are proposed:

H4: The frequency of an industrial product manager's boundary spanning is related:

H4a: Positively to job satisfaction

H4b: Positively to job-related tension

H4c: Positively to performance.

When role conflict and ambiguity reach certain levels, these forces are likely to impinge on the morale of product managers. Product managers may feel anxious about the pressures being faced and the lack of consensus on the functions or duties they must perform. The result may be disillusioned product managers susceptible to burnout. Research by organizational theorists over the last two decades tends to confirm the relationship between role conflict, dissatisfaction, anxiety, propensity to quit and performance [39, 53, 55-57, 73]. Similarly, several marketing studies have found negative relationships between salespersons' role conflict and both job satisfaction and performance [5, 8, 16, 32, 70]. Researchers in both organizational behavior and marketing have also found a definite relationship between role ambiguity and feelings of tension, futility, dissatisfaction, employee turnover and negative attitudes toward role senders [31, 40, 53, 67].

These role pressures are also likely to have an impact on performance. Several recent studies in sales research have found negative relationships between both role conflict and ambiguity with performance [5, 15, 21, 25, 34, 75]. Given this background the following hypotheses are presented:

H5: An industrial product manager's role conflict (dimensions) is related:

H5a: Negatively to job satisfaction

H5b: Positively to job-related tension

H5c: Negatively to performance.

H6: An industrial product manager's role ambiguity is related:

H6a: Negatively to job satisfaction

H6b: Positively to job-related tension

H6c: Negatively to performance.

One business adage is that a happy worker is a more productive one. Bateman and Organ [6], Motowildo [58] and Smith et al. [71] have provided empirical evidence showing a linkage between job satisfaction and prosocial organizational behaviors such as cooperating, listening, volunteering and helping others. According to Cohen's [17] review of several key studies, stress creates conditions of information overload resulting in cognitive fatigue; as a result, energy needed for task performance is sapped. Too much tension can be an insidious force operating to frustrate a product manager and ultimately leading to reduced performance from dramatic exhaustion and physical and mental fatigue [54, 59].

This model follows the thinking of Lazarus et al. [46], Janis and Leventhal [36] and Motowildo et al. [59], who view stress as an intervening variable with antecedent causes and behavioral consequences. The dysfunctional consequences to the firm may be grave in that a tense manager may be prone to more mistakes and may use adverse coping mechanisms. Moreover, the anxiety, hostility and depression which stress may generate are likely to affect job performance with respect to interpersonal sensitivity, tolerance for frustration and overall quality of work. It is therefore hypothesized that:

H7: An industrial product manager's job satisfaction is related positively to performance

H8: An industrial product manager's job-related tension is related negatively to performance.

Methodology

The Sample

One researcher [50] has advocated the use of a single-industry or one organization sample selection of product managers to control for confounding results because of structural differences between industries. In light of this recommendation, the senior management of NZ Telecom was requested and agreed to participate in the study.

The data used to test the research hypotheses were obtained from questionnaires administered in October 1987 to 95 managers employed by NZ Telecom, the former government telecommunications monopoly in New Zealand, which is being privatized. As the monopoly is being deregulated in the late 1980s, profit making is becoming a chief concern for the organization. The respondents were assured that their individual responses would be kept strictly confidential.

The resulting sample size consisted of 69 industrial product managers, all of whom were men. The median age was 37. Approximately 80% had at least some college education. Median job tenure was 15 years as a manager in engineering and/or marketing.

The full-time position of product manager was created at NZ Telecom about 14 months prior to the time of the study; the product managers included in the study had been in their positions 8 to 14 months at the time of the study and most (58) of them reported being involved directly in planning and implementing new product market introductions during the past six months. All the product managers participating in the study reported that product introductions by new competitors, customer inquiry and demand for new products from NZ Telecom, and/or the dramatic increase in the rate of new product launches by NZ Telecom between 1985 to late 1987, were significant influences on the increase in communications between line and planning divisions in the organization as well as in the evaluations of their individual performance levels.

The results of the study may be unique and limited in two ways: they are industry-specific and they are related only to modeling and testing the performance of product managers working in an organization moving from a less competitive environment with few new product introductions

to an intensely competitive environment with many new product introductions.

The Variables

Environmental uncertainty. Since many aspects of uncertainty seem to work through an awareness and perception process [20, 33, 47], perceived environmental uncertainty was used to assess the product manager's evaluation of the environment. There are, as yet, no definitive answers to the appropriateness of a subjective measure versus objective measures of uncertainty. Anderson and Paine [4, p. 821] maintain: "It is generally accepted that the perceptions of environmental and internal characteristics (rather than objective characteristics of the environment) are the important properties to consider." The instrument, using 15 interval scaled items, was adapted from a scale developed by Duncan [23], who identified the main dimensions of uncertainty in decision making. In a study by Downey et al. [20], an alpha coefficient of .67 was reported; hence, the Duncan uncertainty scale "appears to be methodologically adequate for basic research" [20, p. 202].

Boundary spanning. Boundary spanning, defined as the frequency of verbal, written, formal and informal communication beyond the product manager's department, was measured using a modified instrument designed by Aiken and Hage [2] and used successfully by Leifer [48] and Lysonski [50]. Respondents in this study reported the frequencies of various boundary activities excluding strictly social contact on continuous scales; these were aggregated to develop a global measure.

Role pressures. Role conflict is defined as the degree of incompatibility or incongruity of expectations associated with a role. Ten Likert scaled items were used to measure role conflict along various dimensions. The specific types of role conflict measured were:

- person role: incompatibility between the product manager's expectations and the expectations otherwise associated with his role;
- intrasender: incompatible expectations from another person;

- intersender: expectations of one person incompatible with those of another person; and
- role overload: a product manager's responsibilities exceed the time and resources available for accomplishment.

Role ambiguity, defined as a lack of clarity in terms of how one will be evaluated and what is expected, was measured with six-item Likert scale items. Both of these scales were developed and validated by Rizzo et al. [67] while others [69] have reported the psychometric soundness of the scales.

Job satisfaction. Job satisfaction was measured with five Likert type items. The questions assessed the product manager's evaluations about promotion opportunities, chances to demonstrate individual ability and initiative, satisfaction with the position compared to other positions, whether the product manager would recommend the position to others and overall job satisfaction. A scale of this type was used successfully in several other studies [e.g., 7, 66].

Job-related tension. Job-related tension was measured by 11 items employing Likert scales which incorporated measures of work-related worry, relief when leaving work, items from the Manifest Anxiety Scale developed by Taylor [72] and dimensions from a study by House and Rizzo [31]. Job tension is the extent to which an individual is psychologically bothered by work-related matters.

Job performance. Two aspects of performance were measured using two self-rating items. The first item measured overall performance in terms of managing the assigned product line while the second item asked for the product manager's rating in terms of his interpersonal relations in gaining the cooperation of others who can assist in implementing product plans. A ten-point scale was used with endpoints of "just satisfactory" and "clear excellence." Because substantial disagreement surrounds what constitutes a valid objective measure of effective product management, the self-rating scale was ideally suited to this study. Moreover, self-rating measures of performance have been found to have less leniency, less restriction of range and less error than the purportedly more objective measures of superior ratings [30]. Pym and Auld [65] have found self-ratings to correlate strongly with more

Table 1. Reliability of Model Variables

Construct	Number of scale items	Alpha reliabilities
Environmental uncertainty (perceived)	15	.84
Boundary spanning	11	.69
Role conflict	10	.97
Person role	2	
Intersender	4	
Intrasender	2	
Role overload	2	
Role ambiguity	6	.95
Job satisfaction	5	.73
Job-related tension	11	.96
Performance		
Overall	1	
Coordinating others	1	

objective measures of performance, while Pruden and Reese [64] found such measures to be related to actual job performance of industrial salesmen. A review of the literature on self-rated measures of performance by Busch and Bush [15] found strong support for their use. Michaels et al. [52] used this approach in the industrial setting for industrial buyers while Behrman and Perreault [8] used it for industrial salespeople.

Scale Reliabilities

The instruments were used successfully by a number of other researchers. Scale reliabilities (Cronbach alpha coefficients) were derived to determine the internal consistency of the instruments. Table 1 presents these reliability measures while Table 2 presents other scale information for the sample. Because reliabilities exceeding .50 are generally viewed as sufficient for research purposes [60], all scales used in this study can be considered reliable.

Method of Analysis

An appropriate methodology for analyzing the causal properties of the model is path analysis, also referred to as structural equation analysis [43, 62]. From a data analysis viewpoint, a path analysis is simply a series of regression analyses that allows quantification of each linkage in the model in terms of standardized (beta) coefficients that are called path coefficients [10, 22]. Each

Table 2. Correlations, Means and Standard Deviations of Research Variables

Variables	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X _{10a}	X _{10b}
Uncertainty (perceived)	X ₁										
Boundary spanning	X ₂	-.145									
Person role conflict	X ₃	.256	-.072								
Intersender conflict	X ₄	.179	-.111	.355							
Intrasender conflict	X ₅	.287	-.068	.486	.449						
Role overload conflict	X ₆	.101	.112	.251	.419	.602					
Role ambiguity	X ₇	.330	-.295	.363	.301	.372	.195				
Job satisfaction	X ₈	.145	.283	-.213	-.152	-.268	-.087	-.474			
Job-related tension	X ₉	.161	-.034	.258	.404	.344	.431	.120	-.017		
Performance (overall)	X _{10a}	-.344	.058	-.117	-.231	-.142	-.126	-.449	.276	-.185	
Performance (coordinating)	X _{10b}	-.276	.302	-.201	-.361	-.321	-.182	-.305	.165	-.391	.577
Means	74	67.9	7.6	14.4	8.1	7.5	21.0	19.1	33.7	4.9	2.0
Standard deviations	13.8	47.2	2.4	3.7	2.4	2.7	5.4	4.8	8.6	6.0	1.8

endogenous variable was regressed on others that preceded it in the model in Figure 1. Table 3 provides the structural equations of the model. This procedure assumes a general model in which variables that precede particular variables in the causal sequence have direct causal effects on it. The maximum likelihood approach (such as implemented with LISREL) was deemed unsuitable

for testing the model given the sample size and sizable number of constructs.

Figures 2 and 3 present estimates of computed path coefficients. Linkage assessments were not based on the criterion of statistical significance since this is not an accepted method of determining causal significance [22]. Researchers suggest that a path coefficient must show a magnitude of

Table 3. Structural equations

$$\begin{aligned}
 X_2 &= B_{21}X_1 + e_2 \\
 X_3 &= B_{32}X_2 + B_{31}X_1 + e_3 \\
 X_4 &= B_{42}X_2 + B_{41}X_1 + e_4 \\
 X_5 &= B_{52}X_2 + B_{51}X_1 + e_5 \\
 X_6 &= B_{62}X_2 + B_{61}X_1 + e_6 \\
 X_7 &= B_{72}X_2 + B_{71}X_1 + e_7 \\
 X_8 &= B_{81}X_1 + B_{82}X_2 + B_{83}X_3 + B_{84}X_4 + B_{85}X_5 + B_{86}X_6 + B_{87}X_7 + e_8 \\
 X_9 &= B_{91}X_1 + B_{92}X_2 + B_{93}X_3 + B_{94}X_4 + B_{95}X_5 + B_{96}X_6 + B_{97}X_7 + e_9 \\
 X_{10} &= B_{101}X_1 + B_{102}X_2 + B_{103}X_3 + B_{104}X_4 + B_{105}X_5 + B_{106}X_6 + B_{107}X_7 + B_{108}X_8 + B_{109}X_9 + e_{10}
 \end{aligned}$$

Where: B_{ij} = Standardized partial regression coefficients which represent path coefficients

e_i = error term

P_{ij} = path coefficients (i = dependent variable; j = independent variable)

X₁: Perceived uncertainty; X₂: Boundary spanning; X₃: Person role conflict; X₄: Intrasender conflict; X₅: Intersender conflict; X₆: Role overload conflict; X₇: Role ambiguity; X₈: Job satisfaction; X₉: Job-related tension; X_{10a}: Perceived performance (overall); X_{10b}: Perceived performance (coordinating others)

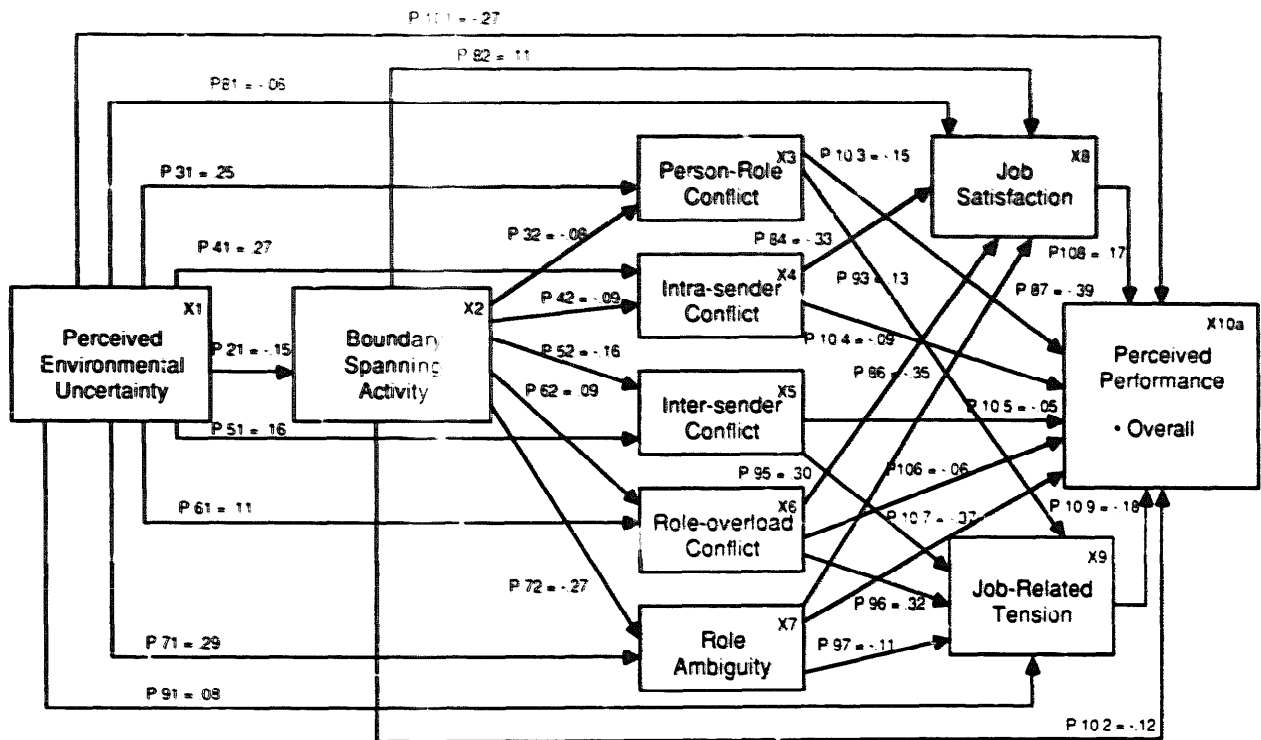
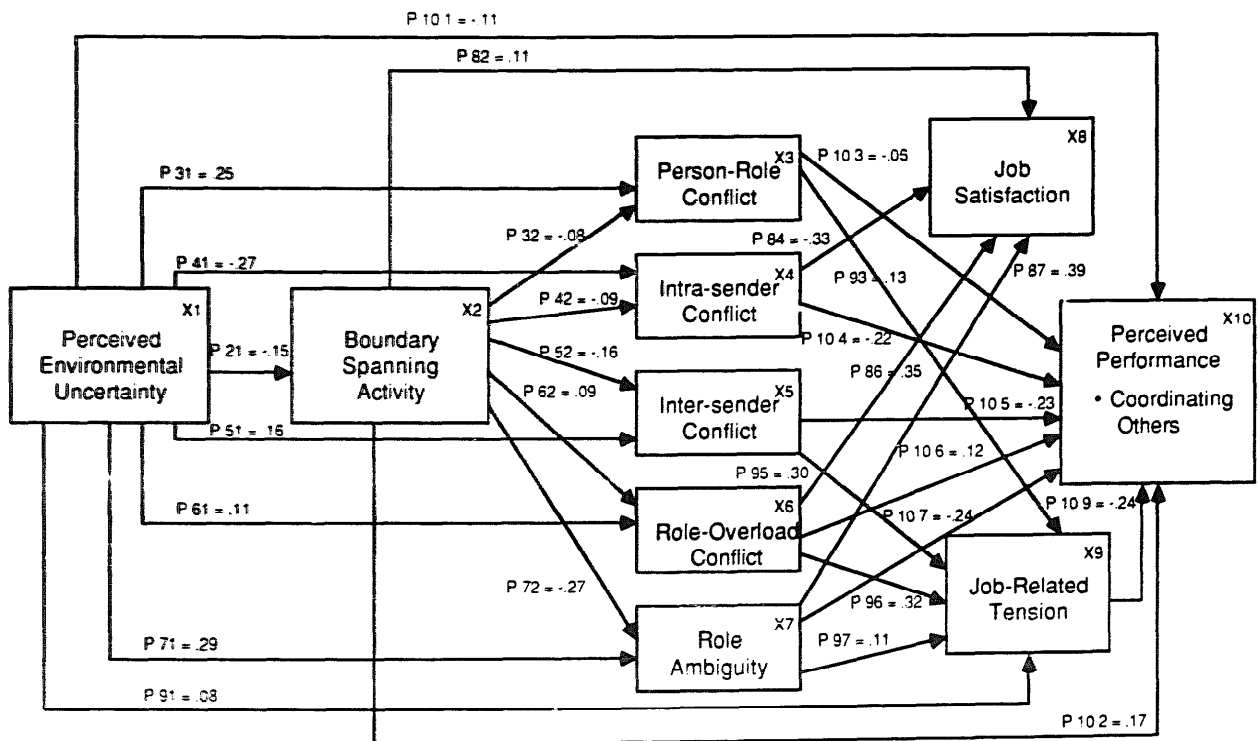


Figure 2. Overall performance as the final dependent variable.

Figure 3. Performance in coordinating others as the final dependent variable.



.05 or greater to be considered causally significant [45]. If any path coefficient was found to be causally nonsignificant, the path was deleted and the system evaluated again.

The basic test for the adequacy of a causal model is to compare the original correlation matrix with the reproduced correlation matrix [29, 43]. The reproduced correlation matrix is calculated from the estimates of the computed path coefficients. In almost every case in which the reproduced correlation matrix was compared to the original one, the discrepancy was less than .06; this suggests that the original model is a meaningful description of the relationships among the variables.

Each performance variable was treated as the final dependent variable in separate models. The models were trimmed by deleting paths with insignificant path coefficients. This produced restricted models in which the insignificant paths identified earlier were assumed to have path coefficients equal to zero. Given this assumption, the remaining path coefficients in the trimmed or restricted models were recalculated by regressing each endogenous variable on only those antecedent variables that were not presumed to have direct paths to it.

In testing the hypothesized models, which specifies direct and indirect linkages, with path analysis some standard assumptions were made. First, the relationships between the variables are assumed to be linear. Examinations of scatter plots of the residuals indicated the residuals were randomly dispersed and the relationships in the study were most likely linear. Moreover, the fact that the scatter plots indicated no serious problem with "outliers" suggested that the model had no misspecification problems. The results are limited to the extent that some variables are omitted from the model specification in the study. Second, it is assumed that the residuals of a given endogenous variable are not correlated with other system variables nor other variables. Third, the causal system posited by the mode is viewed as recursive (i.e., there is no reciprocal causation) [78]. Lastly, the variables are assumed to be measured on interval scales. To the extent that the scales are ordinal, the results may be slightly biased, although multiple regression is still the recommended procedure [11, 35].

Results

Figures 2 and 3 include the path coefficients for the two trimmed models using the two performance measures. Table 4 provides a listing of each path coefficient for the full and trimmed models and the association R^2 s. The results do not uphold H1 in that the perceived environmental uncertainty link to boundary spanning is negative ($p = -.15$). Hence, boundary spanning seems to decrease as uncertainty increases. Given the highly uncertain environment being experienced by NZ Telecom during the time of the study, some of the product managers may have "given up" and retreated into their own organization.¹

H2a was supported in that perceived uncertainty was related to all measures of role conflict, particularly to person role conflict ($p = .25$), intrasender conflict ($p = .27$) and role ambiguity ($p = .29$), supporting H2b. Uncertainty was linked slightly both to satisfaction ($p = .05$), leading to acceptance of H2c and to tension ($p = .08$), lending support to H2d. Although uncertainty was related to performance in coordinating others ($p = -.12$), it was linked more strongly to overall performance ($p = -.27$), suggesting that greater uncertainty hampered performance (as hypothesized in H2e).

Boundary spanning was not strongly associated with the measures of role conflict, although the results indicate it was inversely related to all forms of conflict except role overload conflict; hence, H3a was not supported in the direction hypothesized. H3b is rejected for the same reason in that the relationship between boundary spanning and role ambiguity was negative ($p = -.27$).

Boundary spanning was found to be weakly linked to satisfaction ($p = .14$), supporting H4a, but H4b was rejected since the relationship between boundary spanning and tension was negative and weak ($p = -.04$). Boundary spanning was associated negatively with overall performance ($p = -.12$) and positively with performance in coordinating others ($p = .19$), although these linkages are moderate; H4c was given some support as a result.

¹ One of the journal reviewers of an earlier draft of this article offered this observation on the observed negative relationship.

Table 4. Path Analysis for the Full and Trimmed Models

Dependent variable	Predictor variable	Hypothesized sign	Fully recursive model		Trimmed model	
			Path coefficient	R ²	Path coefficient	R ²
X ₂	X ₁	+	-.15	.03		Same
X ₃	X ₁	+	.25	.07		Same
	X ₂	+	-.08			
X ₄	X ₁	+	.27	.09		Same
	X ₂	+	-.09			
X ₅	X ₁	+	.16	.06		Same
	X ₂	+	-.16			
X ₆	X ₁	+	.11	.02		Same
	X ₂	+	.09			
X ₇	X ₁	+	.29	.18		Same
	X ₂	+	-.27			
X ₈	X ₁	-	-.05	.35	-.06	.33
	X ₂	+	.14		.11	
	X ₃	-	.03			
	X ₄	-	-.28		-.33	
	X ₅	-	.02			
	X ₆	-	-.35		.35	
	X ₇	-	-.43		-.39	
X ₉	X ₁	+	.07	.31	.08	.30
	X ₂	+	-.04		.11	
	X ₃	+	.12		.13	
	X ₄	+	.03		-.33	
	X ₅	+	.29		.30	
	X ₆	+	.31		.32	
	X ₇	+	-.14		-.11	
X _{10a}	X ₁	-	-.27	.31		Same
	X ₂	+	-.12			
	X ₃	-	-.15			
	X ₄	-	-.09			
	X ₅	-	-.05			
	X ₆	-	-.06			
	X ₇	-	-.37			
	X ₈	+	.17			
	X ₉	-	-.18			
X _{10b}	X ₁	-	-.11	.34	-.12	.36
	X ₂	+	.17		.19	
	X ₃	-	-.06		-.07	
	X ₄	-	-.22			
	X ₅	-	-.23		-.22	
	X ₆	-	-.12		-.14	
	X ₇	-	-.24	-.21		
	X ₈	+	.02			
	X ₉	-	-.24	-.28		

X₁: Perceived uncertainty; X₂: Boundary spanning; X₃: Person role conflict; X₄: Intrasender conflict; X₅: Intersender conflict; X₆: Role overload conflict; X₇: Role ambiguity; X₈: Job satisfaction; X₉: Job-related tension; X_{10A}: Perceived performance (overall); X_{10B}: Perceived performance (coordinating others)

H5a was upheld in that two dimensions of role conflict (intrasender with $p = -.33$ and role overload with $p = -.35$) were strongly linked to job satisfaction. The paths for person role and intersender conflict were deleted in Figures 2 and

3 as a result given that their path coefficients were less than .05. H5b was also given support in that three dimensions of role conflict (except intrasender) were linked strongly to job-related tension. The two hypotheses positing a linkage of role

ambiguity to job satisfaction (H6a) and job-related tension (H6b) are also given support with linkages of $p = -.39$ for satisfaction and $p = -.11$ for tension.

The final set of hypotheses examined the paths between role pressures and role outcomes on performance. The hypothesized negative impact of role conflict on performance (H5c) was supported in that all the conflict measures were related inversely to both performance measures. Furthermore, role ambiguity was also negatively associated with performance in coordinating others ($p = -.21$) and with overall performance ($p = -.37$), which supports H6c.

Lastly, job satisfaction was linked to overall performance ($p = .17$) but not to performance involved with coordinating others ($p = .02$), which lends some support to H7. H8 was supported in that tension was linked negatively to both overall performance ($p = -.18$) and to performance involved in coordinating others ($p = .28$). The R^2 s for the model with overall performance as the final dependent variable was .31 and .36 for the coordinating performance measure.

Discussion

Although these results pertain to one organization which recently implemented a product management organization, the findings may be relevant for large organizations studying the question of whether or not to adopt a product management design or to those firms who plan to initiate the system. The results from testing the models show that antecedent variables such as environmental uncertainty and role conflict among others can lead to impaired performance on the part of the industrial product manager. Because so little empirical research of this kind has been carried out on industrial product managers, a path analysis was developed with the assumption that the variables are ordered in this causal sequence.

The results support many of the important links implied by the causal order. However, the interpretations of these path analyses must be tempered with some caution. Among the possible limitations are inability to rule out other plausible causal configurations, incomplete specification of the model may confound interpretation, errors of measurement, correlated method variance among the self-report measures and other potential

Table 5. Comparisons Between Consumer and Industrial Product Managers

Average number of Boundary spanning contacts	Consumer ^a	Industrial ^b
Internal	57.7	21.4*
External	19.9	11.2*
Informal	43.6	20.7*
Formal	23.9	11.3*

^a Source: Lysonski [50].

^b Source: Present study.

*t-test comparisons significant at .01 level.

sources of bias. Despite these limitations, this study represents a step forward in developing causal models on industrial product management.

Several managerial and research implications follow from the study. First, the results indicate that perceived environmental uncertainty may not stimulate greater levels of boundary spanning. Indeed, product managers who did more boundary spanning perceived the environment to be more certain. This finding may be due to the newness of the product management position for most of the NZ Telecom product managers. Table 5 provides a comparison between findings by Lysonski [50] on the boundary spanning activities of consumer product managers and those in this study. On every dimension of boundary spanning, consumer product managers had at least twice as many contacts. Because the consumer setting is different, we must be cautious in interpreting this comparison.

Many other reasons may account for this result. Exhibit 1 lists some of the responses by product managers when asked, Are you satisfied with your level of contacts with the market? Many product managers lamented their lack of time to perform such activities. Our interviews with respondents also suggest that because NZ Telecom historically did not have to deal with competition, managers were unclear how to react to the uncertainty of a deregulated environment.

In addition, some of the respondents' informal comments after completing the questionnaire expressed concern about their authority and responsibilities as product managers. Several reported being told by their senior managers not to engage in boundary spanning actions. One product manager reported the following as a verbatim quote made by the manager he reported to:

Exhibit 1. Selected Candid Remarks by Respondents When Asked, Are You Satisfied with Your Level of Contacts with the Market?

I simply don't have the time to talk with outside clients as much as I'd like to.

Sure it would be nice to meet personally with the market, but I'm busy with matters here [at the head office]; besides, I'm expected to be in my office.

I seldom ask sales reps for their views. We hear about problems when they happen.

I feel overwhelmed at times. I'm supposed to know about the latest development in switching technology and how these can be used in our networks. I have enough trouble making sure the technology we have is on the go.

I wish I had more time to absorb what's happening. With deregulation, we're all finding our way.

The sales reps are supposed to let us know what's happening.

My superior knows the market.

"Your job is to stay in your office and do product planning; operations does the implementing, you do the planning. Period." These directions did not fit the written job description given to each product manager at NZ Telecom. The written job description stated that the product manager is responsible for planning, ensuring that implementation of the plan occurs and measuring results on sales and profits of the implemented plan.

The newness of the job and confusion found among the product managers and their supervisors were factors likely to cause low confidence in performing the actions called for in the job description. Such conditions are likely to intervene and cause a negative relationship between environmental uncertainty and boundary spanning activities. This finding may suggest better training and preparation of product managers before they formally begin their positions. Training programs could be introduced to help product managers better deal with situations of high uncertainty. In addition, senior management may wish to emphasize the need for boundary spanning.

As NZ Telecom experiences more turbulence from operating in a deregulated environment, keeping in touch with the many forms of uncertainty is mandatory for product line viability. One product manager interviewed in the study explained that the situation in his job was rather "cozy" when NZ Telecom was a government monopoly. Such product managers may not be accustomed to dealing with uncertainty and the concomitant need for boundary spanning. However, industrial product managers likely need to be active boundary spanners. In the

words of Aldrich and Herker [3, p. 229] "an organization's ability to adapt to environmental contingencies depends in part on the expertise of the boundary role incumbent in selection, transmitting and interpreting information originating in the environment." Moreover, Jemison [38] argues that boundary spanners are important in strategic decision making because of their ability to recognize and deal with trends of changes in the environment. The implication, therefore, is clear. Industrial product managers are likely to need to span boundaries actively to understand external forces to be able to anticipate changes.

The second implication deals with the impact of uncertainty on the product manager. Perceived uncertainty was related to role pressures of conflict and ambiguity; these variables may lead to adverse outcomes such as dissatisfaction and tension. Uncertainty was also found to be associated with lower performance, particularly overall performance. Hence, product managers, particularly those recently hired, may need better training and skills to cope with and understand uncertainty.

A third implication applies to the benefits of boundary spanning. Unlike many other studies, this study did not find a positive association between role pressures of conflict and ambiguity with boundary spanning. Instead, the results indicated that as boundary spanning increased, these role pressures actually declined. Our interviews suggest that the product managers' efforts to communicate, despite the lack of encouragement, allowed them to understand the expectations of others, and possibly to shape the expectations of others, and, thus, lessening the likelihood of role

conflict. Such a result can also be explained in that boundary spanning activity would better assist the product managers in performing their jobs more effectively. Encouraging product managers to be better boundary spanners may actually help to lessen role conflict and ambiguity.

The product manager's boundary spanning was associated positively with performance in coordinating others but negatively with overall performance. Such a result may stem from the product manager using boundary spanning for coordination but not for environmental scanning, which might help to improve overall product line performance.

A fourth implication focuses on the potentially deleterious effects of role pressures. Most of the dimensions of role conflict were linked to tension and dissatisfaction. Role ambiguity also had a definite negative impact on satisfaction but less so on tension. It is likely that role ambiguity will increase for these product managers as the company confronts competition from deregulation. Role ambiguity is likely to arise as the product managers adapt to the new environment and as senior management begins to redefine their job requirements. Indeed, they may not know what to do in such a changing environment, a type which they never had experienced before. This finding suggests that senior management may need to provide the information necessary for product managers to perform their tasks effectively. Such information may be particularly important for firms such as NZ Telecom that confront increasing competition and advancing technologies.

Performance was clearly seen to be related to various dimensions of role conflict. For example, the conflict experienced by the product manager from a single (intrasender) or from multiple (intersender) role senders was related to diminished performance. This may imply that product managers need to be given some help in their role in dealing with the expectations of others. Consumer product managers often do not have authority commensurate with their responsibility. Role ambiguity also was linked to diminished overall performance. Establishing clearer expectations in terms of how product managers will be evaluated and exactly what is expected might increase their performances.

Methods to mitigate role pressures are many. Kahn et al. [39] suggest several methods includ-

ing changes in the organizational structure, development of new criteria for selection and placement of personnel, increasing the individual's tolerance and coping abilities and strengthening the interpersonal bonds among organizational members.

A fifth implication deals with the effects of satisfaction and tension on performance. A product manager's satisfaction seemed to increase performance, which confirms some of the relationships established in prior studies examining other organizational personnel. Moreover, tension was also linked to reduced performance both in overall terms and in coordinating the actions of others.

Since some forms of stress are to some degree inevitable, the key may be to reduce them to manageable levels [44]. In the present study perceived environmental uncertainty was related to forms of role conflict and ambiguity. It is, therefore, likely that as environmental uncertainty increases, these role pressures will increase.

Because the components of the study are new to research on industrial product management, it is hoped that this study will stimulate further thinking and research on industrial product management. If we are to better understand behavior and performance of product managers, more research on existing product management organizations is needed. Research is also needed to investigate how product managers can be given the requisite skills to deal more with uncertainty, role pressures and an increasing need to communicate. An examination of the moderating effects of a product manager's need for clarity, need for achievement and other personal characteristics on the relationships in the model would also contribute to our understanding. In particular, the development of more integrative models similar to the one used in this study is needed for better understanding of the industrial product management system. Longitudinal studies would also help to better understand another dimension of the relationships in this study.

The various insights and implications derived from this study should be accepted as tentative rather than conclusive because the data analyzed came from a single firm operating in one industry. Therefore, generalizations of the findings must await confirmation through additional studies in-

volving a variety of industrial product management settings.

References

1. Adams, J. Stacey. The structure and dynamics of behavior in organizational boundary roles. In: *Handbook of Organizational Psychology*, M. D. Dunnette (ed.). Chicago: Rand McNally, 1976, pp. 1175-1199.
2. Aiken, Michael and Hage, Gerald. Organizational permeability, boundary spanners and organizational structure. Paper presented at the American Sociological Association Conference, New Orleans, 1972.
3. Aldrich, Howard and Herker, Diane. Boundary spanning roles and organizational structure. *Academy of Management Review* 2:217-230 (1977).
4. Anderson, C. and Paine, F. Managerial perceptions of strategic behavior. *Academy of Management Journal* 18:811-823 (1975).
5. Bagozzi, Richard P. Salesperson performance and satisfaction as a function of individual differences, interpersonal and situational factors. *Journal of Marketing Research* 15:517-531 (1978).
6. Bateman, T. S. and Organ, D. W. Job satisfaction and the good soldier: the relationship between effect and employee citizenship. *Academy of Management Journal* 26:587-595 (1983).
7. Beehr, Terry A., Walsh, Jeffrey T. and Taber, Thomas D. Relationship of stress to individually and organizationally valued states: higher order needs as a moderator. *Journal of Applied Psychology* 61:41-47 (1976).
8. Behrman, Douglas N. and Perreault, William D., Jr. A role stress model of the performance and satisfaction of industrial salespersons. *Journal of Marketing* 48:9-21 (1984).
9. Benson, F. *The Relaxation Response*. New York: William Morrow, 1975.
10. Billings, R. S. and Wroten, S. P. Use of path analysis in industrial/organizational psychology: criticisms and suggestions. *Journal of Applied Psychology* 63:677-688 (1978).
11. Bohrnstedt, G. W. and Carten, T. M. Robustness in regression analysis. In: *Sociological Methodology*, H. L. Coster (ed.). San Francisco: Jossey Bass, 1971, pp. 118-146.
12. Boulton, William R., Lindsay, William M., Franklin, Stephen G. and Rue, Leslie. Strategic planning: determining the impact of environmental characteristics and uncertainty. *Academy of Management Journal* 25:500-509 (1982).
13. Brown, W. B. Systems boundaries and information flow. *Academy of Management Journal* 13:318-327 (1966).
14. Buell, V. P. The changing role of the product manager in consumer goods companies. *Journal of Marketing* 39:3-11 (1975).
15. Busch, P. and Bush, R. F. Women contrasted to men in the industrial salesforce: job satisfaction, values, role clarity, performance and propensity to leave. *Journal of Marketing Research* 15:438-448 (1978).
16. Churchill, Gilbert A., Ford, Neil M. and Walker, Orville C. Measuring the job satisfaction of industrial salesmen. *Journal of Marketing Research* 11:254-260 (1974).
17. Cohen, S. After effects of stress on human performance and social behaviour: a review of research and theory. *Psychological Bulletin* 88:82-108 (1980).
18. Conrath, David W. Organizing decision making behavior under varying conditions of uncertainty. *Management Science* 13:B487-B500 (1967).
19. Cummings, W. T., Jackson, D. W. and Ostrom, L. L. Differences between industrial and consumer product managers. *Industrial Marketing Management* 13:171-180 (1984).
20. Downey, H. Kirk, Hellriegel, Ron and Slocum, John. The effects of individual differences on managers' perceptions of environmental uncertainty. *Proceedings of the 35th Meeting of the Academy of Management*, 1975, 200-202.
21. Dubinsky, A. J. and Mattson, B. E. Consequences of role conflict and ambiguity experienced by retail salespeople. *Journal of Retailing* 55:70-80 (1979).
22. Duncan, Otis D. Path analysis: sociological examples. In: *Causal Models in the Social Sciences*, H. M. Blalock, Jr. (ed.). New York: Macmillan, 1971, 115-138.
23. Duncan, Robert B. Characteristics of organizational environments and perceived environmental uncertainty. *Administrative Science Quarterly* 17:313-327 (1972).
24. Eckles, R. W. and Novotny, T. J. Industrial product manager: authority and responsibility. *Industrial Marketing Management* 13:71-75 (1984).
25. Franke, G. R., Behrman, D. N. and Perreault, W. D. Salesforce performance and satisfaction: contemporaneous relationships and selected antecedents. 1982 AMA Educators' Conference Proceedings. Chicago: American Marketing Association, 1982, 233-237.
26. Fry, Louis W., Futrell, Charles, Parasuraman, A. and Chmielewski, Margaret. An analysis of alternative causal models of salesperson role perceptions and work-related attitudes. *Journal of Marketing Research* 23:153-163 (1986).
27. Galbraith, J. *Organizational Design*. Reading, MA: Addison, 1977.
28. Gemmill, G. R. and Wilemon, D. L. The product manager as an influence agent. *Journal of Marketing* 36:65-73 (1972).
29. Heise, D. R. Problems in path analysis and causal influence. In: *Sociological Methodology*, E. F. Borgatta (ed.). San Francisco: Jossey Bass, 1969, 38-73.
30. Heneman, G. Comparisons of self and superior ratings of managerial performance. *Journal of Applied Psychology* 59:638-642 (1974).
31. House, Robert and Rizzo, John. Role conflict and ambiguity as critical variables in a model of organizational behavior. *Organizational Behavior and Human Performance* 7:467-505 (1972).
32. Howell, R. D., Bellenger, D. N. and Wilcox, J. B. Self esteem, role stress and job satisfaction among marketing managers. *Journal of Business Research* 15:71-84 (1987).
33. Huber, George P., O'Connell, Michael J. and Cummings, Larry. Perceived environmental uncertainty: effects of information and structure. *Academy of Management Journal* 18:725-740 (1975).
34. Ivancevich, J. M. and Donnelly, J. H., Jr. A study of role clarity and need for clarity in three occupational groups. *Academy of Management Journal* 17:28-36 (1974).
35. James, L. R., Mulaik, S. A. and Brett, J. M. *Causal Analysis: Assumptions, Models and Data*. Beverly Hills, CA: Sage Publications, 1982.
36. Janis, I. L. and Leventhal, H. Human reactions to stress. In: *Handbook of Personality Theory and Research*, E. F. Borgatta and W. W. Lambert (eds.). Chicago: Rand McNally, 1968, 1041-1085.
37. Jelinek, Mariann. Rethink strategy or perish. *Journal of Product Innovation Management* 1:36-42 (1984).
38. Jemison, D. B. The importance of boundary spanning roles in strategic decision making. *Journal of Management Studies* 21(2):131-152 (1984).
39. Kahn, Robert L., Wolfe, D. M., Quinn, R. P., Snock, J. D. and Rosenthal, R. A. *Organizational Stress*. New York: Wiley, 1964.
40. Keller, Robert T. Role conflict and ambiguity: correlates with job satisfaction and values. *Personnel Psychology* 28:57-64 (1975).

41. Keller, Robert T. and Szilagyi, Andrew. Employee reactions to leader reward behavior. *Academy of Management Journal* 19:619-627 (1976).
42. Kelly, J. P. and Hise, R. T. Industrial and consumer goods product managers are different. *Industrial Marketing Management* 8:325-332 (1979).
43. Kerlinger, F. N. and Pedhazur, E. J. *Multiple Regression in Behavioral Research*. New York: Holt Rinehart and Winston, 1973.
44. Kets de Vries, Manfred. Organizational stress: a call for management action. *Sloan Management Review* 21(1):3-14 (1979).
45. Land, K. C. Principles of path analysis. In: *Sociological Methodology*. San Francisco: Jossey Bass, 1969, 15-27.
46. Lazarus, R. S., Deese, J. and Osler, J. F. The effects of psychological stress upon performance. *Psychological Bulletin* 49:293-316 (1952).
47. Leblebici, Huseyin and Sarancik, Gerald R. Effects of environmental uncertainty on information and decision processes in banks. *Administrative Science Quarterly* 26(4):578-596 (1981).
48. Leifer, Richard Paul. An analysis of the characteristics and functioning of interorganizational boundary spanning personnel. Ph.D. dissertation, Madison: University of Wisconsin (1975).
49. Luck, David. Interfaces of a product manager. *Journal of Marketing* 33:32-36 (1969).
50. Lysonski, Steven. A boundary theory investigation of the product manager's role. *Journal of Marketing* 49:26-40 (1985).
51. McDaniel, Carl and Gray, D. A. The product manager. *California Management Review* 23:143-150 (1980).
52. Michaels, R. E., Day, R. L. and Joachimsthaler, E. F. Role stress among industrial buyers: an integrative model. *Journal of Marketing* 51:28-45 (1987).
53. Miles, Robert H. A comparison of the relative impacts of role perceptions of ambiguity and conflict by role. *Academy of Management Journal* 19(1):25-35 (1976).
54. Mule, Robert H. Organizational boundary roles. In: *Current Concerns in Occupational Stress*. G. L. Cooper and R. Payne (eds.). New York: Wiley, 1976.
55. Miles, Robert H. Role conflict and ambiguity in boundary and internal roles: a field study using role set analysis and panel design. Ph.D. dissertation, Chapel Hill: University of North Carolina (1974).
56. Miles, Robert H. Role requirements as sources of stress in R&D organizations. *Proceedings of the 35th Meeting of the Academy of Management*, 1975, 191-193.
57. Miles, Robert H. and Perreault, William D., Jr. Organizational role conflict: its antecedents and consequences. *Organizational Behavior and Human Performance* 17:19-44 (1976).
58. Motowilo, S. J. Does job satisfaction lead to consideration and personal sensitivity? *Academy of Management Journal* 27:910-915 (1984).
59. Motowilo, S. J., Packard, J. S. and Manning, M. R. Occupational stress: its causes and consequences for job performance. *Journal of Applied Psychology* 71(4):618-629 (1986).
60. Nunnally, Jim C. *Psychometric Theory*. New York: McGraw-Hill, 1967.
61. Organ, Dennis W. Some variables affecting boundary role behavior. *Sociometry* 34:524-537 (1971).
62. Pedhazur, E. J. *Multiple Regression in Behavioral Research*. 2d ed. New York: Holt, Rinehart and Winston, 1982.
63. Petrini, B. F. and Grub, P. D. Product management in high technology defense industrial marketing. *California Management Review* 15:138-146 (1973).
64. Pruden, Henry O. and Reese, R. Interorganizational role-set relations and the performance and satisfaction of industrial salesmen. *Administrative Science Quarterly* 17:601-609 (1972).
65. Pym, D. L. A. and Auld, H. D. The self-rating as a measure of employee satisfactoriness. *Occupational Psychology* 39:103-113 (1965).
66. Quinn, R. P. and Shepard, L. J. The 1972-1973 quality of employment survey: descriptive statistics with comparison data from the 1969-1970 survey of working conditions. Ann Arbor, MI: The Institute for Social Research, 1974.
67. Rizzo, John, House, Robert and Lirtzman, Sidney. Role conflict and ambiguity in complex organization. *Administrative Science Quarterly* 15:1501-1563 (1970).
68. Roos, Leslie, Jr. and Starke, Frederick. Organizational roles. In: *Handbook of Organizational Design*. P. C. Nystrom and W. H. Starbuck (eds.). New York: Oxford Press, 1981.
69. Schuler, Randall S., Aldag, Ramon J. and Brief, Arthur P. Role conflict and ambiguity: a scale analysis. *Organizational Behavior and Human Performance* 16:111-128 (1977).
70. Sheridan, J. E. and Slocum, J. The direction of the causal relationship between job satisfaction and work performance. *Organizational Behavior and Human Performance* 13:159-172 (1974).
71. Smith, C. A., Organ, D. W. and Near, J. P. Organizational citizenship behavior: its nature and antecedents. *Journal of Applied Psychology* 68:653-663 (1983).
72. Taylor, J. A. A personality scale of manifest anxiety. *Journal of Abnormal Social Psychology* 48:285-290 (1953).
73. Teas, R. K. Supervisory behavior, role stress and the job satisfaction of industrial salespeople. *Journal of Marketing Research* 20:84-91 (1983).
74. Thompson, J. D. *Organizations in Action*. New York: McGraw-Hill, 1967.
75. Walker, O. C., Churchill, G. A. and Ford, N. M. Motivation and performance in industrial selling: present knowledge and needed research. *Journal of Marketing Research* 14:156-168 (1977).
76. Wall, James A. and Adams, J. Stacey. Some variables affecting constituents' evaluations of and behavior toward a boundary role occupant. *Organizational Behavior and Human Performance* 11:390-408 (1974).
77. Weed, Stanley E. and Mitchell, Terence R. the role of environmental and behavioral uncertainty as a mediator of situation-performance relationships. *Academy of Management Journal* 23:36-60 (1980).
78. Wright, Sewall. Path coefficients and path regressions: alternative or complementary concepts? *Biometrics* 16:189-202 (1970).